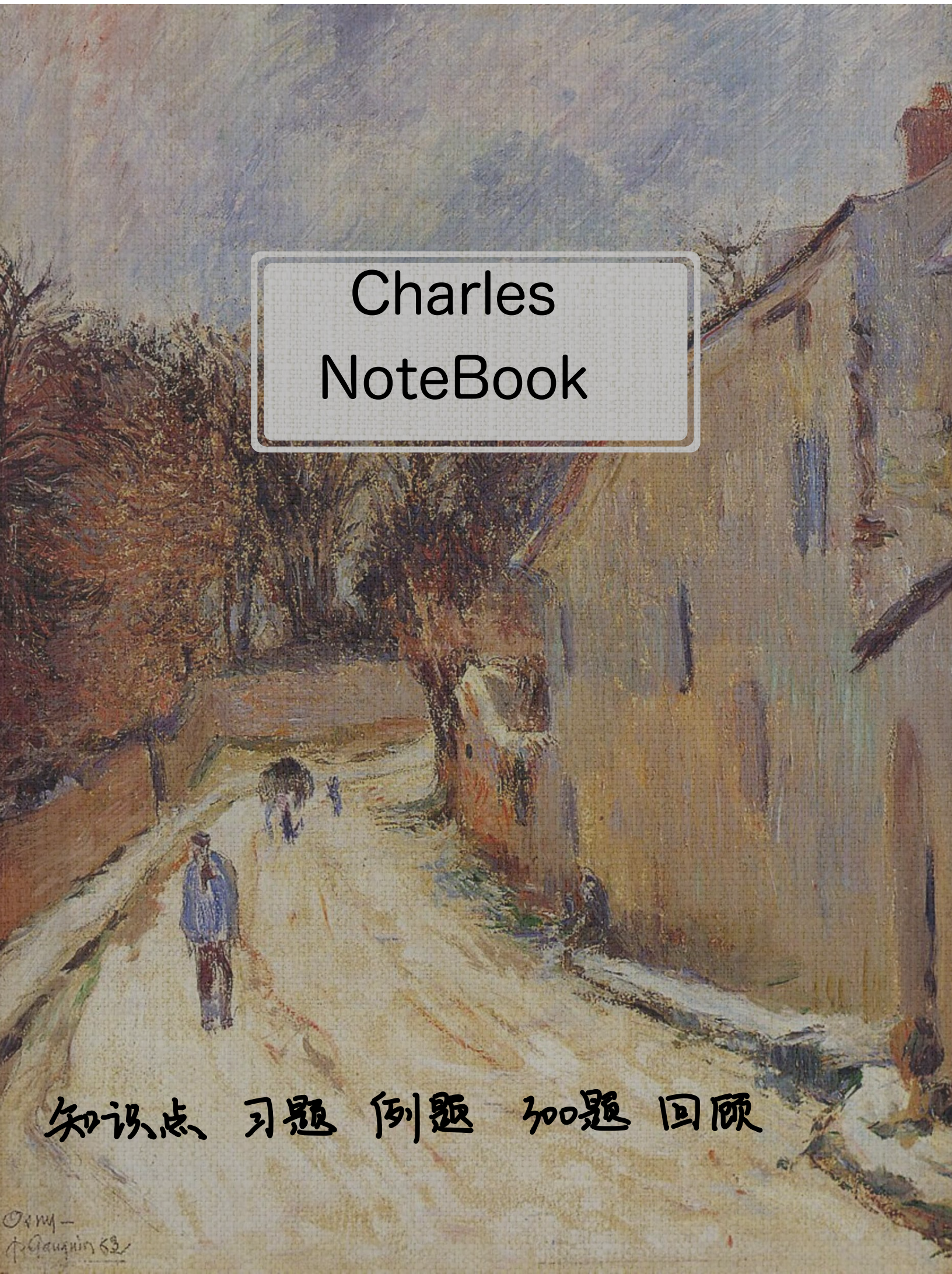


An impressionist painting of a street scene. On the left, there are large, leafy trees in shades of brown and orange. A path leads from the foreground into the distance. Several figures are visible: a person in a blue coat stands in the foreground on the left, and a group of people is further down the path. On the right, a tall, light-colored building with vertical brushstrokes stands. The sky is a mix of blue and white, suggesting a bright, slightly overcast day. The overall style is loose and painterly, characteristic of Impressionism.

Charles NoteBook

知识点 习题 例题 加题 回顾

一. 向量代数

数量积

2. 向量的运算及其应用

设 $a = (a_x, a_y, a_z)$, $b = (b_x, b_y, b_z)$, $c = (c_x, c_y, c_z)$, a, b, c 均是非零向量.

(1) 数量积(内积、点积)及其应用.

$$\textcircled{1} a \cdot b = (a_x, a_y, a_z) \cdot (b_x, b_y, b_z) = a_x b_x + a_y b_y + a_z b_z.$$

$$\textcircled{2} a \cdot b = |a| |b| \cos \theta, \text{ 则 } \cos \theta = \frac{a \cdot b}{|a| |b|} = \frac{a_x b_x + a_y b_y + a_z b_z}{\sqrt{a_x^2 + a_y^2 + a_z^2} \cdot \sqrt{b_x^2 + b_y^2 + b_z^2}}, \text{ 其中 } \theta \text{ 为 } a, b \text{ 的夹角.}$$

$$\textcircled{3} a \perp b \Leftrightarrow \theta = \frac{\pi}{2} \Leftrightarrow a \cdot b = |a| |b| \cos \theta = 0 \Leftrightarrow a_x b_x + a_y b_y + a_z b_z = 0.$$

$$\textcircled{4} \text{Proj}_b a = \frac{a \cdot b}{|b|^2} b = \frac{a_x b_x + a_y b_y + a_z b_z}{b_x^2 + b_y^2 + b_z^2} b, \text{ 称为 } a \text{ 在 } b \text{ 上的投影.}$$

定“位置关系”

向量积

(2) 向量积(外积、叉积)及其应用.

$$\textcircled{1} a \times b = \begin{vmatrix} i & j & k \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}, \text{ 其中 } |a \times b| = |a| |b| \sin \theta, \text{ 用右手规则确定方向(转向角不超过 } \pi), \theta \text{ 为 } a, b \text{ 的夹角.}$$

$$\textcircled{2} a \parallel b \Leftrightarrow \theta = 0 \text{ 或 } \pi \Leftrightarrow \frac{a_x}{b_x} = \frac{a_y}{b_y} = \frac{a_z}{b_z}.$$

混合积

(3) 混合积及其应用.

$$\textcircled{1} [abc] = (a \times b) \cdot c = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix}.$$

$$\textcircled{2} \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix} = 0 \Leftrightarrow \text{三向量共面.}$$

向量的方向角与方向余弦

3. 向量的方向角和方向余弦

(1) 非零向量 a 与 x 轴、 y 轴和 z 轴正向的夹角 α, β, γ 称为 a 的方向角.

$$\textcircled{2} \cos \alpha, \cos \beta, \cos \gamma \text{ 称为 } a \text{ 的方向余弦, 且 } \cos \alpha = \frac{a_x}{|a|}, \cos \beta = \frac{a_y}{|a|}, \cos \gamma = \frac{a_z}{|a|}.$$

$$\textcircled{3} a^0 = \frac{a}{|a|} = (\cos \alpha, \cos \beta, \cos \gamma) \text{ 称为向量 } a \text{ 的单位向量(表示方向的向量).}$$

任意向量 $r = xi + yj + zk = (r \cos \alpha, r \cos \beta, r \cos \gamma) = r(\cos \alpha, \cos \beta, \cos \gamma)$, 其中 $\cos \alpha, \cos \beta, \cos \gamma$ 为 r 的方向余弦, r 为 r 的模, $\cos \alpha = \frac{x}{\sqrt{x^2 + y^2 + z^2}}, \cos \beta = \frac{y}{\sqrt{x^2 + y^2 + z^2}}, \cos \gamma = \frac{z}{\sqrt{x^2 + y^2 + z^2}},$
 $r = \sqrt{x^2 + y^2 + z^2}.$

二. 空间平面与直线

平面方程

1. 平面方程

以下假设平面的法向量 $n = (A, B, C)$.

$$\textcircled{1} \text{一般式: } Ax + By + Cz + D = 0.$$

$$\textcircled{2} \text{点法式: } A(x - x_0) + B(y - y_0) + C(z - z_0) = 0.$$

$$\textcircled{3} \text{三点式: } \begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x - x_2 & y - y_2 & z - z_2 \\ x - x_3 & y - y_3 & z - z_3 \end{vmatrix} = 0 \text{ (平面过不共线的三点 } P_i(x_i, y_i, z_i), i = 1, 2, 3).$$

$$\textcircled{4} \text{截距式: } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \text{ (平面过 } (a, 0, 0), (0, b, 0), (0, 0, c) \text{ 三点).}$$

直线方程

2. 直线方程

以下假设直线的方向向量 $\tau = (l, m, n)$.

①一般式: $\begin{cases} A_1x + B_1y + C_1z + D_1 = 0, n_1 = (A_1, B_1, C_1), \\ A_2x + B_2y + C_2z + D_2 = 0, n_2 = (A_2, B_2, C_2), \end{cases}$ 其中 n_1 不平行于 n_2 .

【注】其几何背景很直观,是两个平面的交线,且该直线的方向向量 $\tau = n_1 \times n_2$.

②点向式(标准式、对称式): $\frac{x-x_0}{l} = \frac{y-y_0}{m} = \frac{z-z_0}{n}$.

③参数式: $\begin{cases} x = x_0 + lt, \\ y = y_0 + mt, P_0(x_0, y_0, z_0) \text{ 为直线上的已知点, } t \text{ 为参数.} \\ z = z_0 + nt, \end{cases}$

④两点式: $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$ (直线过不同的两点 $P_i(x_i, y_i, z_i), i=1, 2$).



位置关系

(1)距离.

点 $P_0(x_0, y_0, z_0)$ 到平面 $Ax + By + Cz + D = 0$ 的距离 $d = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}$.

(2)直线间的关系.

设 $\tau_1 = (l_1, m_1, n_1), \tau_2 = (l_2, m_2, n_2)$ 分别为直线 L_1, L_2 的方向向量.

① $L_1 \perp L_2 \Leftrightarrow \tau_1 \perp \tau_2 \Leftrightarrow l_1l_2 + m_1m_2 + n_1n_2 = 0$.

② $L_1 // L_2 \Leftrightarrow \tau_1 // \tau_2 \Leftrightarrow \frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$.

(3)平面间的关系.

设平面 π_1, π_2 的法向量分别为 $n_1 = (A_1, B_1, C_1), n_2 = (A_2, B_2, C_2)$.

① $\pi_1 \perp \pi_2 \Leftrightarrow n_1 \perp n_2 \Leftrightarrow A_1A_2 + B_1B_2 + C_1C_2 = 0$.

② $\pi_1 // \pi_2 \Leftrightarrow n_1 // n_2 \Leftrightarrow \frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$.

(4)平面与直线的关系.

设直线 L 的方向向量为 $\tau = (l, m, n)$, 平面 π 的法向量为 $n = (A, B, C)$.

① $L \perp \pi \Leftrightarrow \tau // n \Leftrightarrow \frac{A}{l} = \frac{B}{m} = \frac{C}{n}$.

② $L // \pi \Leftrightarrow \tau \perp n \Leftrightarrow Al + Bm + Cn = 0$.

三. 空间曲线与曲面

1. 空间曲线

(1)一般式 $\Gamma: \begin{cases} F(x, y, z) = 0, \\ G(x, y, z) = 0, \end{cases}$ 其几何背景为两个曲面的交线.

(2)参数方程 $\Gamma: \begin{cases} x = \varphi(t), \\ y = \psi(t), t \in [\alpha, \beta], \\ z = \omega(t), \end{cases}$

★(3)空间曲线在坐标面上的投影(重点).

以曲线 Γ 在 xOy 平面上的投影曲线为例. 将 $\Gamma: \begin{cases} F(x, y, z) = 0, \\ G(x, y, z) = 0 \end{cases}$ 中的 z 消去, 得到 $\varphi(x, y) = 0$, 则曲

线 Γ 在 xOy 面上的投影曲线包含于曲线 $\begin{cases} \varphi(x, y) = 0, \\ z = 0. \end{cases}$

★例17.3

求曲线 $\begin{cases} (x+2)^2 - z^2 = 4, \\ (x-2)^2 + y^2 = 4 \end{cases}$ 在 yOz 平面上的投影曲线方程.

$yOz \rightarrow$ 消去 x

$$\begin{cases} x^2 + 4x + 4 - z^2 = 4 \\ x^2 - 4x + 4 + y^2 = 4 \end{cases}$$

$$8x = z^2 + y^2$$

$$x = \frac{z^2 + y^2}{8} \Rightarrow \begin{cases} (y^2 + z^2)^2 + 32(y^2 - z^2) = 0 \\ x = 0 \end{cases}$$

2. 空间曲面

曲面名称	方程	图形	曲面名称	方程	图形
椭球面	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$		椭圆锥面	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$ eg. $x^2 + y^2 = z^2$ $z = \sqrt{x^2 + y^2} \leftarrow$ 双叶	
单叶双曲面	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ + + - 两正一负		双曲抛物面(马鞍面)	$\frac{x^2}{2p} - \frac{y^2}{2q} = z \ (p, q > 0)$	
双叶双曲面	$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ + - - 一正两负			$z = xy$	
椭圆抛物面	$\frac{x^2}{2p} + \frac{y^2}{2q} = z \ (p, q > 0)$				

(3) 柱面: 动直线沿定曲线平行移动所形成的曲面.

椭圆柱面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (当 $a=b$ 时为圆柱面).

双曲柱面 $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

抛物柱面 $y = ax^2$.

【注】在空间解析几何中, 一般认为缺少变量的方程为柱面.

(4) 旋转曲面(重点): 曲线 Γ 绕一条定直线旋转一周所形成的曲面.

曲线 $\Gamma: \begin{cases} F(x, y, z) = 0, \\ G(x, y, z) = 0 \end{cases}$ 绕直线 $L: \frac{x-x_0}{m} = \frac{y-y_0}{n} = \frac{z-z_0}{p}$ 旋转形成一个旋转

曲面, 旋转曲面方程的求法如下.

如图 17-1 所示, 已知点 $M_0(x_0, y_0, z_0)$, 方向向量 $s = (m, n, p)$. 在母线 Γ 上任取一点 $M_1(x_1, y_1, z_1)$, 则过 M_1 的纬圆上任意一点 $P(x, y, z)$ 满足条件

1) $\overline{M_1P} \perp s$, 2) $|\overline{M_0P}| = |\overline{M_0M_1}|$, 3) 在 Γ 上

$$\begin{cases} m(x-x_1) + n(y-y_1) + p(z-z_1) = 0 \\ (x_1-x_0)^2 + (y_1-y_0)^2 + (z_1-z_0)^2 = (x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 \\ F(x_1, y_1, z_1) = 0 \quad G(x_1, y_1, z_1) = 0 \end{cases} \rightarrow \text{消去 } x_1, y_1, z_1 \Rightarrow \text{得 } H(x, y, z) = 0$$

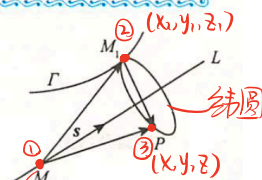


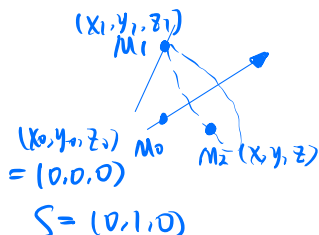
图 17-1

旋转曲面方程

例 17.4

例 17.4

求直线 $L: \begin{cases} x-y+2z-1=0, \\ x-3y-2z+1=0 \end{cases}$ 绕 y 轴旋转一周所形成的曲面方程.



$$1) 0(x-x_1) + (y-y_1) + 0(z-z_1) = 0 \\ \Rightarrow y = y_1 \quad ①$$

$$2) (x_1-0)^2 + (y_1-0)^2 + (z_1-0)^2 = (x-0)^2 + (y-0)^2 + (z-0)^2 \\ \Rightarrow x_1^2 + z_1^2 = x^2 + z^2 \quad ②$$

$$3) \begin{cases} x_1 - y_1 + 2z_1 - 1 = 0 \\ x_1 - 3y_1 - 2z_1 + 1 = 0 \end{cases} \Rightarrow y_1 = x_1 + 2z_1 - 1 \quad ④$$

$$-2x_1 - 8z_1 + 4 = 0$$

$$-x_1 - 4z_1 + 2 = 0 \Rightarrow x_1 = 2 - 4z_1 \quad ③$$

$$\text{由 } ①④ \Rightarrow y = y_1 = 1 - 2z_1 \Rightarrow z_1 = \frac{1-y}{2} \quad ⑤$$

$$\text{由 } ②③ \quad (2-4z_1)^2 + z_1^2 = x^2 + z^2 \quad ⑥$$

$$\text{由 } ①⑥$$

$$16y^2 + (1-y)^2 = 4(x^2 + z^2)$$

$$4x^2 - 17y^2 + 4z^2 + 2y - 1 = 0$$

四 多元函数微分学的几何应用

1. 空间曲线的切线与法平面

(1) 设空间曲线 Γ 由参数方程 $\begin{cases} x = \varphi(t), \\ y = \psi(t), \\ z = \omega(t) \end{cases}$ 给出, 其中 $\varphi(t), \psi(t), \omega(t)$ 均可导, $P_0(x_0, y_0, z_0)$

是 Γ 上对应 $t = t_0$ 的点, 且当 $t = t_0$ 时, $\varphi'(t_0), \psi'(t_0), \omega'(t_0)$ 都不为 0, 则

① 曲线 Γ 在点 $P_0(x_0, y_0, z_0)$ 处的切向量为 $\tau = (\varphi'(t_0), \psi'(t_0), \omega'(t_0))$.

② 曲线 Γ 在点 $P_0(x_0, y_0, z_0)$ 处的切线方程为 $\frac{x-x_0}{\varphi'(t_0)} = \frac{y-y_0}{\psi'(t_0)} = \frac{z-z_0}{\omega'(t_0)}$.

③ 曲线 Γ 在点 $P_0(x_0, y_0, z_0)$ 处的法平面(过点 $P_0(x_0, y_0, z_0)$ 且与切线垂直的平面)方程为

$$\varphi'(t_0)(x-x_0) + \psi'(t_0)(y-y_0) + \omega'(t_0)(z-z_0) = 0.$$

(2) 设空间曲线 Γ 由交面式方程 $\begin{cases} F(x, y, z) = 0, \\ G(x, y, z) = 0 \end{cases}$ 给出, 则在以下表达式有意义的条件下, 有

① 曲线 Γ 在点 $P_0(x_0, y_0, z_0)$ 处的切向量为

$$\tau = \left(\begin{vmatrix} F'_y & F'_z \\ G'_y & G'_z \end{vmatrix}_{P_0}, \begin{vmatrix} F'_x & F'_z \\ G'_x & G'_z \end{vmatrix}_{P_0}, \begin{vmatrix} F'_x & F'_y \\ G'_x & G'_y \end{vmatrix}_{P_0} \right).$$

② 曲线 Γ 在点 $P_0(x_0, y_0, z_0)$ 处的切线方程为

$$\frac{x-x_0}{\begin{vmatrix} F'_y & F'_z \\ G'_y & G'_z \end{vmatrix}_{P_0}} = \frac{y-y_0}{\begin{vmatrix} F'_x & F'_z \\ G'_x & G'_z \end{vmatrix}_{P_0}} = \frac{z-z_0}{\begin{vmatrix} F'_x & F'_y \\ G'_x & G'_y \end{vmatrix}_{P_0}}.$$

③ 曲线 Γ 在点 $P_0(x_0, y_0, z_0)$ 处的法平面方程为

$$\begin{vmatrix} F'_y & F'_z \\ G'_y & G'_z \end{vmatrix}_{P_0} (x-x_0) + \begin{vmatrix} F'_x & F'_z \\ G'_x & G'_z \end{vmatrix}_{P_0} (y-y_0) + \begin{vmatrix} F'_x & F'_y \\ G'_x & G'_y \end{vmatrix}_{P_0} (z-z_0) = 0.$$

2. 空间曲面的切平面与法线

(1) 设空间曲面 Σ 由方程 $F(x, y, z) = 0$ 给出, $F(x, y, z)$ 可微, $P_0(x_0, y_0, z_0)$ 是 Σ 上的点, 则

① 曲面 Σ 在点 $P_0(x_0, y_0, z_0)$ 处的法向量(垂直于该点切平面的向量)为

$$\mathbf{n} = (F'_x(x_0, y_0, z_0), F'_y(x_0, y_0, z_0), F'_z(x_0, y_0, z_0)),$$

且法线方程为

$$\frac{x-x_0}{F'_x(x_0, y_0, z_0)} = \frac{y-y_0}{F'_y(x_0, y_0, z_0)} = \frac{z-z_0}{F'_z(x_0, y_0, z_0)}.$$

② 曲面 Σ 在点 $P_0(x_0, y_0, z_0)$ 处的切平面方程为

$$F'_x(x_0, y_0, z_0)(x-x_0) + F'_y(x_0, y_0, z_0)(y-y_0) + F'_z(x_0, y_0, z_0)(z-z_0) = 0.$$

(2) 设空间曲面 Σ 由方程 $z = f(x, y)$ 给出, $f(x, y)$ 可微, 令 $F(x, y, z) = f(x, y) - z$, $P_0(x_0, y_0, z_0)$ 是 Σ 上的点, 则

① 曲面 Σ 在点 $P_0(x_0, y_0, z_0)$ 处的法向量为

$$\mathbf{n} = (f'_x(x_0, y_0), f'_y(x_0, y_0), -1),$$

且法线方程为

$$\frac{x-x_0}{f'_x(x_0, y_0)} = \frac{y-y_0}{f'_y(x_0, y_0)} = \frac{z-z_0}{-1}.$$

② 曲面 Σ 在点 $P_0(x_0, y_0, z_0)$ 处的切平面方程为

$$f'_x(x_0, y_0)(x-x_0) + f'_y(x_0, y_0)(y-y_0) - (z-z_0) = 0.$$

【注】若用 α, β, γ 表示曲面 $z = f(x, y)$ 在点 (x_0, y_0, z_0) 处的法向量的方向角, 并假定法向量的方向是向上的, 即它与 z 轴正向所成的角 γ 是锐角, 则法向量的方向余弦为

$$\cos \alpha = \frac{-f'_x}{\sqrt{1+(f'_x)^2+(f'_y)^2}}, \quad \cos \beta = \frac{-f'_y}{\sqrt{1+(f'_x)^2+(f'_y)^2}}, \quad \cos \gamma = \frac{1}{\sqrt{1+(f'_x)^2+(f'_y)^2}},$$

其中, $f'_x = f'_x(x_0, y_0)$, $f'_y = f'_y(x_0, y_0)$.

例 17.2

例 17.2 设直线 $L: \begin{cases} x+y+b=0, \\ x+ay-z-3=0 \end{cases}$ 在平面 π 上, 而平面 π 与曲面 $z = x^2 + y^2$ 相切于点 $(1, -2, 5)$, 求

a, b 的值.

$$\text{曲面: } F(x, y, z) = z - x^2 - y^2$$

$$\vec{n} = (F'_x, F'_y, F'_z) = (-2x, -2y, 1)$$

$$\vec{n}(1, -2, 5) = (-2, 4, 1)$$

$$\text{切平面: } -2(x-1) + 4(y+2) + (z-5) = 0 \Rightarrow 2x - 4y - z - 5 = 0$$

$$x+y+b=0 \Rightarrow y = -x-b$$

$$x+ay-z-b=0 \Rightarrow z = x+ay-b = x+a(-x-b)-b$$

$$\Rightarrow \underline{2x + 4x + 4b - x + ax + ab} = 2$$

$$(5+a)=0 \Rightarrow a = -5$$

$$4b + ab = -b = 2 \Rightarrow b = -2$$

例 17.5

例 17.5 求空间曲线 $\Gamma: \begin{cases} x = \int_0^t e^u \cos u \, du, \\ y = 2\sin t + \cos t, \\ z = 1 + e^{3t} \end{cases}$ 在 $t = 0$ 处的切线方程和法平面方程.

$$\begin{aligned}x_t' &= e^t \cos t & x_0 &= 0 \\y_t' &= 2\cos t - \sin t & y_0 &= 1 \\z_t' &= 3e^{3t} & z_0 &= 2 \\T|_{t=0} &= (1, 2, 3) \\切线 & \frac{x-0}{1} = \frac{y-1}{2} = \frac{z-2}{3}\end{aligned}$$

$$\text{法平面 } 1(x-0) + 2(y-1) + 3(z-2) = 0$$

五 场论初步

1. 方向导数

在许多问题中,不仅要知道函数在坐标轴方向上的变化率(即偏导数),还要设法求得函数在某点沿着其他特定方向上的变化率,这就是本部分所要讨论的方向导数.

定义 1 设三元函数 $u=u(x,y,z)$ 在点 $P_0(x_0, y_0, z_0)$ 的某空间邻域 $U \subset \mathbb{R}^3$ 内有定义, l 为从点 P_0 出发的射线, $P(x,y,z)$ 为 l 上且在 U 内的任一点, 则

$$\begin{cases}x-x_0=\Delta x=t\cos\alpha, \\y-y_0=\Delta y=t\cos\beta, \\z-z_0=\Delta z=t\cos\gamma.\end{cases}$$

以 $t=\sqrt{(\Delta x)^2+(\Delta y)^2+(\Delta z)^2}$ 表示 P 与 P_0 之间的距离, 如图 17-2 所示, 若极限

$$\lim_{t \rightarrow 0} \frac{u(P)-u(P_0)}{t} = \lim_{t \rightarrow 0} \frac{u(x_0+t\cos\alpha, y_0+t\cos\beta, z_0+t\cos\gamma)-u(x_0, y_0, z_0)}{t}$$

存在, 则称此极限为函数 $u=u(x,y,z)$ 在点 P_0 沿方向 l 的方向导数, 记作 $\left. \frac{\partial u}{\partial l} \right|_{P_0}$.

定理(方向导数的计算公式) 设三元函数 $u=u(x,y,z)$ 在点 $P_0(x_0, y_0, z_0)$ 处可微分, 则 $u=u(x,y,z)$ 在点 P_0 处沿任一方向 l 的方向导数都存在, 且

$$\left. \frac{\partial u}{\partial l} \right|_{P_0} = u'_x(P_0)\cos\alpha + u'_y(P_0)\cos\beta + u'_z(P_0)\cos\gamma,$$

其中 $\cos\alpha, \cos\beta, \cos\gamma$ 为方向 l 的方向余弦.

【注】 对于二元函数 $f(x,y)$ 的情况与三元函数类似.

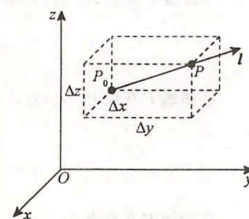


图 17-2

例 17.6

例 17.6 求函数 $z=xe^{2y}$ 在点 $P(1,0)$ 处沿点 $P(1,0)$ 指向点 $Q(2,-1)$ 方向的方向导数.

$$\begin{aligned}\overrightarrow{PQ} &= (1, -1) \\ \cos\alpha &= \frac{1}{\sqrt{1^2+(-1)^2}} = \frac{\sqrt{2}}{2} \\ \cos\beta &= \frac{-1}{\sqrt{1^2+(-1)^2}} = -\frac{\sqrt{2}}{2}\end{aligned}$$

$$\begin{aligned}\left. \frac{\partial z}{\partial l} \right|_{(1,0)} &= z'_x \Big|_{(1,0)} \cdot \cos\alpha + z'_y \Big|_{(1,0)} \cdot \cos\beta \\ &= e^{2y} \Big|_{(1,0)} \cdot \frac{\sqrt{2}}{2} - 2xe^{2y} \Big|_{(1,0)} \cdot \frac{\sqrt{2}}{2} \\ &= \frac{\sqrt{2}}{2} - \frac{2\sqrt{2}}{2} = -\frac{\sqrt{2}}{2}\end{aligned}$$

2 梯度

定义 2 设三元函数 $u=u(x,y,z)$ 在点 $P_0(x_0, y_0, z_0)$ 处具有一阶偏导数, 则定义

$$\text{grad } u \Big|_{P_0} = (u'_x(P_0), u'_y(P_0), u'_z(P_0))$$

为函数 $u=u(x,y,z)$ 在点 P_0 处的梯度.

3. 方向导数与梯度关系

由方向导数的计算公式 $\frac{\partial u}{\partial l} \Big|_{P_0} = u'_x(P_0) \cos \alpha + u'_y(P_0) \cos \beta + u'_z(P_0) \cos \gamma$ 与梯度的定义

$$\text{grad } u \Big|_{P_0} = (u'_x(P_0), u'_y(P_0), u'_z(P_0)),$$

$$\begin{aligned} \text{可得到 } \frac{\partial u}{\partial l} \Big|_{P_0} &= (u'_x(P_0), u'_y(P_0), u'_z(P_0)) \cdot (\cos \alpha, \cos \beta, \cos \gamma) = \text{grad } u \Big|_{P_0} \cdot l^0 \\ &= |\text{grad } u \Big|_{P_0}| |l^0| \cos \theta = |\text{grad } u \Big|_{P_0}| \cos \theta, \end{aligned}$$

其中 θ 为 $\text{grad } u \Big|_{P_0}$ 与 l^0 的夹角, 当 $\cos \theta = 1$ 时, $\frac{\partial u}{\partial l} \Big|_{P_0}$ 有最大值.

结论 函数在某点的梯度是一个向量, 它的方向与取得最大方向导数的方向一致, 而它的模为方向导数的最大值.

4 散度与旋度

设向量场 $A(x,y,z) = (P(x,y,z), Q(x,y,z), R(x,y,z))$, 则

散度

$$\text{div } A = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z},$$

旋度

$$\text{rot } A = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}.$$

例 17.7

例 17.7 函数 $u = \ln(x^2 + y^2 + z^2)$ 在点 $M(1, 2, -2)$ 处的梯度 $\text{grad } u \Big|_M =$ _____.

$$\begin{aligned} u'_x &= \frac{2x}{x^2+y^2+z^2} \Big|_M = \frac{2}{9} \\ u'_y &= \frac{2y}{x^2+y^2+z^2} \Big|_M = \frac{4}{9} \\ u'_z &= \frac{2z}{x^2+y^2+z^2} \Big|_M = \frac{-4}{9} \\ \text{grad } u \Big|_M &= \left(\frac{2}{9}, \frac{4}{9}, \frac{-4}{9} \right) \end{aligned}$$

例 17.8

例 17.8 向量场 $u(x,y,z) = xy^2i + ye^zj + x\ln(1+z^2)k$ 在点 $P(1, 1, 0)$ 处的散度 $\text{div } u =$ _____.

$$\begin{aligned} \text{div } u \Big|_P &= \frac{\partial P}{\partial x} \Big|_P + \frac{\partial Q}{\partial y} \Big|_P + \frac{\partial R}{\partial z} \Big|_P \\ &= y^2 \Big|_P + e^z \Big|_P + \frac{x \cdot 2z}{1+z^2} \Big|_P \\ &= 1 + 1 + 0 = 2 \end{aligned}$$

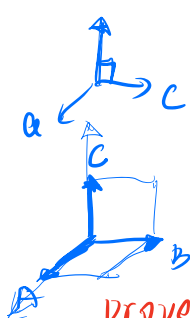
例 17.9

设 $F(x,y,z) = xyi - yzj + zxk$, 则 $\text{rot } F(1, 1, 0) =$ _____.

$$\begin{aligned} \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & -yz & zx \end{vmatrix} &= \left(\frac{\partial zx}{\partial y} + \frac{\partial yz}{\partial z} \right) i + \left(\frac{\partial xy}{\partial z} - \frac{\partial zx}{\partial x} \right) j + \left(-\frac{\partial yz}{\partial x} - \frac{\partial xy}{\partial y} \right) k \\ &= yi - zj - xk \quad \text{rot } F(1, 1, 0) = \vec{i} - \vec{k} \end{aligned}$$

例 17.1

设 $(a \times b) \cdot c = 2$, 则 $[(a+b) \times (b+c)] \cdot (c+a) = \underline{\hspace{2cm}}$.

$$\begin{aligned}
 &= [a \times (b+c) + b \times (b+c)] \cdot (c+a) \\
 &= [a \times b + a \times c + \cancel{b \times b} + b \times c] \cdot (c+a) \\
 &= a \times b \cdot c + \cancel{a \times c \cdot c} + \cancel{b \times c \cdot c} + \cancel{a \times b \cdot a} + \cancel{a \times c \cdot a} + b \times c \cdot a \\
 &= 2a \times b \cdot c = 4?
 \end{aligned}$$


prove $(a \times b) \cdot c = a \cdot (b \times c)$

$$b \times c = \begin{vmatrix} i & j & k \\ b_i & b_j & b_k \\ c_i & c_j & c_k \end{vmatrix} = (b_j c_k - b_k c_j) i + (b_k c_i - b_i c_k) j + (b_i c_j - b_j c_i) k$$

$$b \times c \cdot a = \underline{(a_i b_j c_k - a_i b_k c_j)} + \underline{(a_j b_k c_i - a_j b_i c_k)} + \underline{(a_k b_i c_j - a_k b_j c_i)}$$

$$a \times b = \begin{vmatrix} i & j & k \\ a_i & a_j & a_k \\ b_i & b_j & b_k \end{vmatrix} = (a_j b_k - a_k b_j) i + (a_k b_i - a_i b_k) j + (a_i b_j - a_j b_i) k$$

$$a \times b \cdot c = \underline{(a_j b_k c_i - a_k b_j c_i)} + \underline{(a_k b_i c_j - a_i b_k c_j)} + \underline{(a_i b_j c_k - a_j b_i c_k)}$$

例 17.1

17.1 与两直线

$$\begin{cases} x=1, \\ y=-1+t, \text{ 及 } \frac{x+1}{1} = \frac{y+2}{2} = \frac{z-1}{1} \\ z=2+t \end{cases}$$

都平行, 且过原点的平面方程为 $\underline{\hspace{2cm}}$.

平面 $A(x-0) + B(y-0) + C(z-0) = 0$.



$$T_1 = (0, 1, 1)$$

$$T_2 = (1, 2, 1)$$

$$T = T_1 \times T_2 = \begin{vmatrix} i & j & k \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{vmatrix} = (-1, 1, -1)$$

$$\Rightarrow -x + y - z = 0$$

例 17.2

求旋转抛物面 $z = x^2 + y^2 - 1$ 在点 $(2, 1, 4)$ 处的切平面及法线方程.

$$f(x, y, z) = z - x^2 - y^2 + 1 = 0$$

$$\vec{n} = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \bigg|_{(2, 1, 4)} = (-2x, -2y, 1) \bigg|_{(2, 1, 4)} = (-4, -2, 1)$$

$$\begin{aligned} \text{切平面: } -4(x-2) - 2(y-1) + (z-4) &= 0 \\ -4x - 2y + z + 6 &= 0 \end{aligned}$$

$$\text{法线方程: } \frac{x-2}{-4} = \frac{y-1}{-2} = \frac{z-4}{1}$$

例 17.3

求曲面 $z = e^x + 2xy = 3$ 在点 $(1, 2, 0)$ 处的切平面及法线方程.

$$F(x, y, z) = z - e^x + 2xy - 3 = 0$$

$$\left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right) \bigg|_{(1, 2, 0)} = (2y, 2x, 1 - e^x) \bigg|_{(1, 2, 0)} = (4, 2, 0)$$

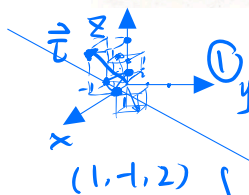
$$\text{切平面 } 4(x-1) + 2(y-2) + 0(z-0) = 0$$

$$\text{法线 } \frac{x-1}{4} = \frac{y-2}{2} = \frac{z-0}{0}$$

例 17.4

17.4 求直线 $l: \frac{x-1}{1} = \frac{y}{1} = \frac{z-1}{-1}$ 在平面 $\pi: x-y+2z-1=0$ 上的投影直线 l_0 的方程, 并求 l_0 绕 y 轴

旋转一周所成曲面的方程.



$$\vec{r} = (1, -1, 2)$$

$$\vec{n} = (1, 1, -1)$$

$$(1, -1, 2)$$

$$\text{令 } u = (1, -1, 2)$$

$$(a, b, 1)(1, -1, 2) = 0$$

$$\Rightarrow a - b + 2 = 0 \Rightarrow \text{令 } a = -1 \text{ 则 } b = 1$$

$$\Rightarrow \text{令 } v = (-1, 1, 1)$$

$$\text{令 } w = \begin{vmatrix} i & j & k \\ 1 & -1 & 2 \\ -1 & 1 & 1 \end{vmatrix} = (-3, -3, 0)$$

$$\Rightarrow \text{则 } \begin{cases} u = (1, -1, 2) \\ v = (-1, 1, 1) \\ w = (-3, -3, 0) \end{cases}$$

$$(x, y, z) \rightarrow (u, v, w)$$

$$\begin{cases} x = (\frac{1}{6}, \frac{2}{6}, \frac{1}{6}) \\ y = (\frac{-1}{6}, \frac{2}{6}, \frac{1}{6}) \\ z = (1, 1, 0) \end{cases}$$

$$(u, v, w) \rightarrow (x, y, z)$$

$$\text{直线: } \begin{cases} x-1=y \\ z-1=1-x \end{cases}$$

$$\Rightarrow \begin{cases} u-2v-3=0 \\ 7u+4v-w-12=0 \end{cases}$$

$$\text{消去 } u \Rightarrow \begin{cases} 18v-w+9=0 \\ u=1 \end{cases}$$

$$(x, y, z) \Rightarrow \begin{cases} -5x+7y+6z+3=0 \\ x-y+2z-1=0 \end{cases}$$



$$\textcircled{2} \quad \text{Proj}_{\vec{n}} \vec{m} = |\vec{n}| \cos \theta = \frac{2}{\sqrt{6}} = \frac{\sqrt{6}}{3}$$

$$\vec{m} = \frac{\sqrt{6}}{3} \frac{(1, -1, 2)}{\sqrt{6}} = \left(\frac{1}{3}, -\frac{1}{3}, \frac{2}{3}\right)$$

$$\vec{l} = \vec{n} - \vec{m} = \left(\frac{-4}{3}, -\frac{2}{3}, \frac{1}{3}\right) \quad \text{and } \vec{l} =$$

$$\left\{ \begin{array}{l} \frac{x-1}{1} = \frac{y}{1} = \frac{z-1}{-1} \\ x-y+2z-1=0 \end{array} \right\} \quad \text{交点: } (2, 1, 0)$$

$$\Rightarrow l_0: \frac{x-2}{4} = \frac{y-1}{2} = \frac{z-0}{1}$$

$$\textcircled{3} \quad \begin{array}{ll} 1) \text{ y轴一点} & P_0 (0, 0, 0) \\ 2) \text{ l_0上一点} & P_1 (x_0, y_0, z_0) \\ 3) & P (x, y, z) \end{array} \left\{ \begin{array}{l} |P_1 P_0| = |PP_0| \Rightarrow x_0^2 + y_0^2 + z_0^2 = x^2 + y^2 + z^2 \quad \textcircled{1} \\ P_1 P \perp \vec{l} \Rightarrow y = y_0 \quad \textcircled{2} \\ (x - x_0, y - y_0, z - z_0) \cdot (0, 1, 0) = 0 \end{array} \right.$$

$$4) P_1 \text{ 在 } l \text{ 上: } \frac{x_0-2}{4} = \frac{y_0-1}{2} = \frac{z_0}{1}$$

$$\Rightarrow \left. \begin{array}{l} x_0 = 2y_0 \\ y_0 = 2z_0 + 1 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} x_0 = 2y_0 = 2y \\ y_0 = y_0 = y \\ z_0 = \frac{1}{2}(y_0 - 1) = \frac{1}{2}(y - 1) \end{array} \right.$$

$$4y^2 + y^2 + \frac{1}{4}(y^2 + 1 - 2y) = x^2 + y^2 + z^2$$

$$(6y^2 + y^2 + 1 - 2y) = z^2 + x^2$$

习 17.5

17.5 函数 $u = \ln(x + \sqrt{y^2 + z^2})$ 在点 $A(1, 0, 1)$ 处沿点 A 指向点 $B(3, -2, 2)$ 方向的方向导数为_____.

$$\vec{AB} = (2, -2, 1)$$

$$\cos \alpha = \frac{2}{3}$$

$$\cos \beta = \frac{-2}{3}$$

$$\cos \gamma = \frac{1}{3}$$

$$u'_x = \frac{1}{x + \sqrt{y^2 + z^2}}$$

$$u'_y = \frac{y \cdot (y^2 + z^2)^{-\frac{1}{2}}}{x + \sqrt{y^2 + z^2}}$$

$$u'_z = \frac{z \cdot (y^2 + z^2)^{-\frac{1}{2}}}{x + \sqrt{y^2 + z^2}}$$

$$u'_x \cos \alpha + u'_y \cos \beta + u'_z \cos \gamma \Big|_{(1, 0, 1)}$$

$$= \frac{1}{2} \times \frac{2}{3} + 0 + \frac{1}{2} \times \frac{1}{3}$$

$$= \frac{1}{2}$$

300.17.1

设 $|a+b| = |a-b|$, 且 $a = (3, -5, 8)$, $b = (-1, 1, z)$, 则 $z =$ _____.

$$a+b = (2, -4, 8+z)$$

$$a-b = (4, -6, 8-z)$$

$$4+16+64+z^2+16z = 16+36+64+z^2-16z$$

$$32z = 32 \Rightarrow z = 1$$

300.17.2

过点 $(1, 2, -1)$ 且过 z 轴的平面方程为



$$\vec{n} = \begin{vmatrix} i & j & k \\ 0 & 0 & 1 \\ 1 & 2 & -1 \end{vmatrix} = (-2, 1, 0) \Rightarrow -2x + y = 0$$

300.17.3

过点 $(1, 2, -1)$ 且垂直于平面 $2x - y + 3z = 5$ 的直线方程为



$$\vec{n} = (2, -1, 3)$$

$$\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z+1}{3}$$

300.17.4

17.4 设直线 $l: \begin{cases} x+y-z+1=0, \\ x-y+3z+3=0, \end{cases}$ 平面 $\Pi: x-2y-z+3=0$, 则直线 l (

(A) 平行于 Π

(B) 在 Π 上

(C) 垂直于 Π

(D) 与 Π 相交但不垂直

方向向量: $\left(\begin{vmatrix} \frac{\partial F}{\partial y} & \frac{\partial F}{\partial z} \\ \frac{\partial G}{\partial y} & \frac{\partial G}{\partial z} \end{vmatrix}, \begin{vmatrix} \frac{\partial F}{\partial z} & \frac{\partial F}{\partial x} \\ \frac{\partial G}{\partial z} & \frac{\partial G}{\partial x} \end{vmatrix}, \begin{vmatrix} \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} \\ \frac{\partial G}{\partial x} & \frac{\partial G}{\partial y} \end{vmatrix} \right)$

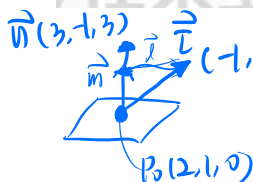
$$= \left(\begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix}, \begin{vmatrix} -1 & 1 \\ 3 & 1 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} \right)$$

$$= (2, -4, -2)$$

(C)

300.17.5

求直线 $l: \frac{x-1}{1} = \frac{y}{1} = \frac{z-1}{-1}$ 在平面 $\Pi: 3x - y + 3z = 5$ 上的投影直线 l_0 的方程.



$$\vec{n} = (3, -1, 3)$$

$$\vec{\tau} = (-1, -1, 1)$$

$$\left\{ \frac{x-1}{1} = \frac{y}{1} = \frac{z-1}{-1} \right\} \Rightarrow P_0(2, 1, 0)$$

$$3x - y + 3z = 5$$

$$\text{proj}_{\Pi} \vec{\tau} = \frac{\vec{n} \cdot \vec{\tau}}{|\vec{n}|} = \frac{1}{\sqrt{19}}$$

$$\vec{m} = \frac{(3, -1, 3)}{\sqrt{19}} \cdot \frac{1}{\sqrt{19}} = \frac{1}{19} (3, -1, 3)$$

$$\vec{l} = \left(\frac{-22}{19}, \frac{-18}{19}, \frac{16}{19} \right) \rightarrow a \vec{l} = (-11, -9, 8)$$

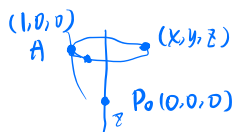
$$\text{do: } \frac{x-2}{-11} = \frac{y-1}{-9} = \frac{z}{8}$$

300.17.6

经过点 $A(1,0,0)$ 与点 $B(0,1,1)$ 的直线绕 z 轴旋转一周生成的曲面方程是

$$\vec{l} = (-1, 1, 1)$$

$$l: \frac{x-1}{-1} = \frac{y}{1} = \frac{z}{1}$$



$$\left. \begin{array}{l} ① \quad x_0^2 + y_0^2 + z_0^2 = x^2 + y^2 + z^2 \\ ② \quad z = z_0 \\ ③ \quad \frac{x_0-1}{-1} = y_0 = z_0 \end{array} \right\}$$

$$x_0 = -z_0$$

$$\begin{aligned} 1^2 + z^2 - 2z + 2z^2 &= x^2 + y^2 + z^2 \\ x^2 + y^2 - 2z^2 + 2z - 1 &= 0 \end{aligned}$$

300.17.7

17.7 在曲线 $x=t, y=-t^2, z=t^3$ 的所有切线中, 与平面 $x+2y+z=4$ 平行的切线()。

(A) 只有 1 条 (B) 只有 2 条 (C) 至少有 3 条 (D) 不存在

$$\begin{cases} x=t \\ y=-t^2 \\ z=t^3 \end{cases}$$

$$\vec{r}(1, -2t, 3t^2)$$

$$\vec{r}_1(1, 2, 1)$$

$$1 - 4t + 3t^3 = 3t^3 - 4t + 1 = 0$$

$$\Delta = 16 - 12 = 4 > 0 \quad (B)$$

300.17.8

已知 $F = x^3 i + y^3 j + z^3 k$, 则在点 $(1, 0, -1)$ 处的 $\text{div} F$ 为

$$\text{div} F = 3x^2 + 3y^2 + 3z^2 \Big|_{(1,0,-1)} = 6$$

300.17.9

设 $u = x^2 + 3y + yz$, 则 $\text{div}(\text{grad } u) =$

$$\text{grad } u = (2x, 3+y, y)$$

$$\text{div}(\text{grad } u) = 2 + 0 + 0 = 2$$

300.17.10

向量场 $A = (z, 3x, 2y)$ 在点 $M(x, y, z)$ 处的旋度 $\text{rot } A =$

$$\text{rot } A = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & 3x & 2y \end{vmatrix} = 2i + 1j + 3k$$

300.17.11

17.11 已知曲面 $z = 4 - x^2 - y^2$ 上点 P 处的切平面平行于平面 $2x + 2y + z - 1 = 0$, 则点 P 的坐标是

(A) $(1, -1, 2)$ (B) $(-1, 1, 2)$ (C) $(1, 1, 2)$ (D) $(-1, -1, 2)$

$$F(x, y, z) = z + x^2 + y^2 - 4 = 0$$

$$\vec{n} = (2x, 2y, 1)$$

$$\vec{n}_1 = (2, 2, 1)$$

$$P(1, 1, 2) \quad (C)$$

300.17.12

17.12 设 u 是由方程 $e^{xz} - xy - yz - zu = 0$ 所确定的 x, y, z 的隐函数, 求 $u = u(x, y, z)$ 在点 $P(1, 1, 0)$ 处方向导数的最大值.

$\frac{u}{e^z}$
17.12

$$F(x, y, z, u) = e^{xz} - xy - yz - zu = 0$$

$$F'_x|_P = -y|_P = -1$$

$$F'_y|_P = -x - z|_P = 1$$

$$F'_z|_P = e^{xz} - y - u|_P = 0$$

$$F'_u|_P = e^{xz} - z|_P = 1$$

$$\frac{\partial u}{\partial x}|_P = \frac{-F'_x}{F'_u}|_P = \frac{y}{e^{xz} - z}|_P = 1$$

$$\frac{\partial u}{\partial y}|_P = \frac{-F'_y}{F'_u}|_P = \frac{x + z}{e^{xz} - z}|_P = -1$$

$$\frac{\partial u}{\partial z}|_P = \frac{-F'_z}{F'_u}|_P = \frac{-e^{xz} + y + z}{e^{xz} - z}|_P = 0$$

$$\max = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$